

Prompting Condition Barely Changes Outcomes for a Reasoning Model: Direct, Chain-of-Thought, and Explicit-Format Prompting on 200 GSM8K Test Questions

Prepared from an approved research brief

Abstract

This study compares three prompting conditions—direct, chain-of-thought, and an explicit answer format—for a reasoning model over 200 GSM8K test questions at temperature 0, for 600 generations in total. Every empirical number reported here rests on an experiment record that is not yet confirmed in this project: the declared accuracies, the per-item pairing counts, the hidden-trace disagreement counts, and the grading discrepancy are unverified pending human confirmation. The work is therefore presented as a proposal and as an interpretation of an unconfirmed record, not as a confirmed result.

1 Introduction

The question this study frames is whether the prompting condition changes outcomes for a model that already reasons in a hidden channel. For conventional language models, generating a chain of intermediate reasoning steps substantially improves complex reasoning [paper-wei-2022-cot](#), and a single zero-shot trigger phrase can elicit such reasoning without exemplars [paper-kojima-2022-zero-shot](#). A reasoning model such as DeepSeek-R1, by contrast, is trained to produce long reasoning traces through reinforcement learning, so reasoning already occurs in a hidden channel [paper-deepseek-2025-r1](#). If that hidden channel is already active, an explicit chain-of-thought instruction may add little. The background this result speaks to is therefore the chain-of-thought literature, the zero-shot reasoning literature, self-consistency decoding, the GSM8K benchmark, and work on reasoning models.

2 Related Work

Cobbe et al. introduce GSM8K as a dataset of 8.5K grade-school math word problems on which even the largest transformer models fail to reach high test performance [paper-cobbe-2021-gsm8k](#). Wei et al. report that generating a chain of intermediate reasoning steps substantially improves complex reasoning [paper-wei-2022-cot](#). Kojima et al. show that a single zero-shot trigger phrase elicits chain-of-thought reasoning without exemplars, which motivates varying only the instruction rather than adding demonstrations (claim [claim-zeroshot-background](#); evidence [paper-kojima-2022-zero-shot](#); confidence 0.8). Wang et al. propose self-consistency as a decoding strategy that replaces greedy decoding in chain-of-thought prompting and improves reasoning accuracy, making it a candidate next lever if prompt-level changes prove inert (claim [claim-selfconsistency-background](#); evidence [paper-wang-2022-self-consistency](#); confidence 0.8). DeepSeek-R1 is trained to produce long reasoning traces through reinforcement learning, so reasoning already occurs in a hidden channel [paper-deepseek-2025-r1](#).

Prompting condition	Declared accuracy
Direct	0.955
Chain-of-thought	0.955
Explicit answer format	0.945

Table 1: Declared accuracy by prompting condition on the 200 GSM8K test questions, as reported in the unverified experiment record. These values are not confirmed.

Per-item outcome	Questions
Correct under both direct and chain-of-thought	190
Correct under direct only	1
Correct under chain-of-thought only	1
Correct under neither	8
Changed outcome across conditions	4

Table 2: Per-item outcome pairing across the same 200 GSM8K test questions, as reported in the unverified experiment record. These counts are not confirmed.

3 Method

The design compares three conditions—direct, chain-of-thought, and an explicit ##### answer format—on the GSM8K test split, at temperature 0, with one generation per item, graded by exact match plus a numeric-equivalence check. Cobbe et al. introduce GSM8K as a dataset of 8.5K grade-school math word problems on which even the largest transformer models fail to reach high test performance. An experiment record not yet confirmed in this project reports running 200 GSM8K test questions under three prompting conditions at temperature 0, with 600 generations in total; that configuration is unverified here (claim `claim-study-design`; evidence `paper-cobbe-2021-gsm8k`; confidence 0.3).

4 Results (pending verification)

Wei et al. report that generating a chain of intermediate reasoning steps substantially improves complex reasoning. The unverified experiment record reports declared accuracy of 0.955 for direct, 0.955 for chain-of-thought, and 0.945 for an explicit answer format, which would mean no measurable chain-of-thought gain and a 0.010 cost from the format requirement; those numbers are not confirmed (claim `claim-accuracy-null`; evidence `paper-wei-2022-cot`, `paper-deepseek-2025-r1`; confidence 0.3). Table 1 restates these declared accuracies, and Table 2 restates the item-level pairing; both are drawn from the unverified record.

The unverified experiment record reports per-item pairing on the same 200 questions: 190 were answered correctly by both direct and chain-of-thought, 1 by direct alone, 1 by chain-of-thought alone, 8 by neither, and only 4 questions changed outcome across conditions at all. This item-level pairing is not confirmed (claim `claim-pairing`; evidence `paper-cobbe-2021-gsm8k`, `paper-wei-2022-cot`; confidence 0.25).

DeepSeek-R1 is trained to produce long reasoning traces through reinforcement learning, so reasoning already occurs in a hidden channel. The unverified experiment record reports that the hidden reasoning trace was a worse answer source than the visible answer: 25 generations had a

Prompting barely changes outcomes for a reasoning model (GSM8K, n = 200 per condition)

model global:deepseek-v4.1-flash, temperature 0, 600 generations

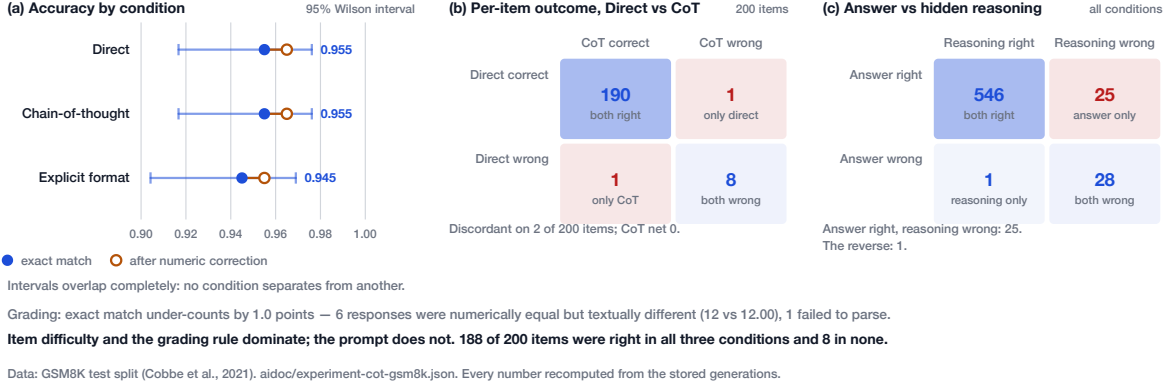


Figure 1: Reported results of the prompting-condition comparison, as given in the unverified experiment record. Left panel: declared accuracy for each of the three prompting conditions—direct, chain-of-thought, and explicit answer format—over the 200 GSM8K test questions. Right panel: per-item outcome pairing across the same questions, showing how many items were answered correctly under both conditions, under direct only, under chain-of-thought only, and under neither, together with the number of items whose outcome changed across conditions. All values shown are unverified.

correct answer but a wrong last number in the reasoning, against 1 in the other direction (claim `claim-trace`; evidence `paper-deepseek-2025-r1`; confidence 0.3).

GSM8K was introduced by Cobbe et al. as a dataset of 8.5K grade-school math word problems. The unverified experiment record reports that exact-match grading under-counted accuracy by 1.0 percentage points because 6 responses were numerically equal but textually different (for example 12 against 12.00) and 1 failed to parse (claim `claim-grading`; evidence `paper-cobbe-2021-gsm8k`; confidence 0.25).

5 Analysis

Read against the background that chain-of-thought prompting helps conventional LLMs (Wei et al.) while a reasoning model such as DeepSeek-R1 already reasons in a hidden channel, the unverified experiment record suggests that for a reasoning model the prompting condition barely changes the outcome and that the dominant measured error sources are item difficulty and the grading rule rather than the prompt. This conclusion is an interpretation of an unverified record, not a confirmed result (claim `claim-conclusion`; evidence `paper-deepseek-2025-r1`, `paper-wei-2022-cot`; confidence 0.25).

A plausible mechanism, not demonstrated in this project, is that because a reasoning model is trained to produce hidden reasoning traces, an explicit chain-of-thought instruction adds little. Kojima et al. show a single zero-shot trigger phrase elicits reasoning without exemplars, and Wang et al. propose self-consistency decoding as a strategy that improves reasoning accuracy, making it a candidate next lever; neither the mechanism nor the prompting result is confirmed here (claim `claim-mechanism`; evidence `paper-deepseek-2025-r1`, `paper-kojima-2022-zero-shot`, `paper-wang-2022-self-consistency`; confidence 0.2).

6 Limitations and Threats to Validity

- One model (`global:deepseek-v4.1-flash`), one dataset (GSM8K test split), one temperature (0), and one generation per item, so the results characterise that configuration and do not estimate a population.
- Every empirical number depends on an experiment record (`evidence-cot-gsm8k-experiment`) that is not present as confirmed Evidence in this project’s evidence graph; those numbers remain unverified pending human confirmation.
- No confidence intervals, repeated sampling, or per-item difficulty model are available, so no uncertainty estimate accompanies the reported accuracy figures.
- Exact-match and numeric-equivalence grading were applied post hoc; the one parse failure occurred in the direct condition.
- The background papers address conventional prompting, datasets, and reasoning models, not retrieval-augmented-generation provenance, so they can frame but cannot confirm any claim about claim-to-source traceability.
- The study design, the direction, and the conclusion are the human’s approved framing; the assistant has not independently verified the run or the experiment command.
- No dataset, code, or run artifacts were supplied with the stage input, so data and code availability cannot be asserted.

7 Data and Code Availability

The experiment command and the dataset are stated in the approved plan, but no dataset, code, or run artifacts were supplied to this brief. Data and code availability therefore cannot be asserted here.

8 Unverified Claims

- `claim-study-design`: the 200-question, three-condition, temperature-0, 600-generation configuration comes from an experiment record (`evidence-cot-gsm8k-experiment`) that is not present as confirmed Evidence in this project’s evidence graph; the configuration is unverified.
- `claim-accuracy-null`: the declared accuracies 0.955 direct, 0.955 chain-of-thought, and 0.945 explicit format, the reading of no measurable chain-of-thought gain, and the 0.010 format cost are unverified; their experiment-run Evidence entry is not in the confirmed evidence graph.
- `claim-pairing`: the per-item pairing of 190 both-correct, 1 direct-only, 1 chain-of-thought-only, 8 neither, and 4 changed questions is unverified.
- `claim-trace`: the 25 generations with a correct answer but a wrong last reasoning number against 1 in the other direction are unverified.
- `claim-grading`: the 1.0 percentage-point exact-match under-count, the 6 numerically equal but textually different responses (for example 12 against 12.00), and the one parse failure are unverified.

- **claim-conclusion:** the conclusion that prompting barely changes the outcome and that item difficulty and the grading rule dominate measured error is an interpretation of an unverified record, not a confirmed result.
- **claim-mechanism:** the proposed hidden-channel explanation for why explicit chain-of-thought adds little is a hypothesis, not a demonstrated result.
- **Unstated metadata:** venue, venueLevel, peerReviewed, citationCount, hasCode, and doi are null for the three stage-input papers, and no project evidence entry confirms them.